

Lab: Variational Calculus and the Free Energy Functional

Learning Goal: Derive, decompose, and analyze variational free energy through mathematical exercises.

Part 1: Derivation Exercise

Exercise: Starting from the definition of model evidence, derive the variational free energy bound step by step.

Given: A generative model $p(o, s) = p(o|s)p(s)$, observations o , hidden states s , and approximate posterior $q(s)$.

1. Write the log model evidence: $\ln p(o) = \ln \int p(o, s) ds$
2. Multiply and divide by $q(s)$ inside the integral
3. Apply Jensen's inequality to obtain the ELBO
4. Define $F[q] = -\text{ELBO}$ and show $F[q] = E_q[-\ln p(o, s)] + E_q[\ln q(s)]$
5. Show $F[q] = -\ln p(o) + D_{\text{KL}}[q(s) \parallel p(s|o)]$
6. Conclude that $F[q] \geq -\ln p(o)$ with equality iff $q(s) = p(s|o)$

Write your response here

Part 2: Decomposition Analysis

Learning Goal: Work through all three decompositions and understand their computational significance.

Exercise: For a simple model — $p(s) = N(0, \sigma^2_p)$, $p(o|s) = N(s, \sigma^2_o)$, and $q(s) = N(\mu_q, \sigma^2_q)$ — compute each decomposition analytically:

Decomposition 1 (Energy - Entropy):

- Energy = $E_q[-\ln p(o, s)] = ?$
- Entropy = $H[q] = \frac{1}{2} \ln(2\pi e \sigma^2_q)$
- $F = \text{Energy} - \text{Entropy}$

Decomposition 2 (Complexity - Accuracy):

- Complexity = $D_{\text{KL}}[q(s) \parallel p(s)] = ?$
- Accuracy = $E_q[\ln p(o|s)] = ?$
- $F = \text{Complexity} - \text{Accuracy}$

Decomposition 3 (Surprise + KL):

- Compute $p(o)$ analytically for this Gaussian model
- Surprise = $-\ln p(o) = ?$
- $KL = D_{\text{KL}}[q(s) \parallel p(s|o)] = ?$
- Verify $F = \text{Surprise} + KL$

Write your response here

Part 3: Mean-Field Updates

Learning Goal: Derive and apply mean-field variational updates.

Exercise: Consider a model with two hidden state groups s_1 and s_2 :

$$p(o, s_1, s_2) = p(o \mid s_1, s_2) \cdot p(s_1) \cdot p(s_2)$$

Assume mean-field factorization: $q(s_1, s_2) = q_1(s_1) \cdot q_2(s_2)$

1. Derive the optimal $q_1^*(s_1)$ by taking the functional derivative of F with respect to q_1 and setting to zero
2. Show that $\ln q_1^*(s_1) = E_{q_2}[\ln p(o, s_1, s_2)] + \text{const}$

3. Explain why this creates an iterative scheme (q_1 depends on q_2 and vice versa)
4. Under what conditions does this scheme converge? (Are there guarantees?)

Write your response here

Part 4: Free Energy on a Factor Graph

Learning Goal: Decompose free energy into local contributions on a simple factor graph.

Exercise: Consider a Hidden Markov Model with 3 time steps:

$$p(s_1, s_2, s_3, o_1, o_2, o_3) = p(s_1) \cdot p(s_2|s_1) \cdot p(s_3|s_2) \cdot p(o_1|s_1) \cdot p(o_2|s_2) \cdot p(o_3|s_3)$$

1. Draw the factor graph (variable nodes and factor nodes)
2. Write the total free energy as a sum of local free energies (one per factor node)
3. What messages need to pass between variable and factor nodes?
4. How does updating beliefs at one node affect neighboring nodes?
5. What is the computational advantage of this decomposition vs. computing F globally?

Write your response here

Part 5: Approximation Comparison

Learning Goal: Compare variational approximation methods.

Exercise: For a bimodal posterior $p(s|o)$ (two well-separated modes), compare:

Method	Approximation	What It Gets Right	What It Gets Wrong
Mean-field (factorized Gaussian)	Single Gaussian	Captures one mode location	Misses the other mode entirely (mode-seeking)
Laplace	Gaussian at MAP	Captures the MAP mode accurately	Misses second mode; underestimates uncertainty
Bethe	Pairwise interactions	Better for structured models	Still struggles with multimodality
MCMC (for comparison)	Samples from true posterior	Both modes captured	Computationally expensive; convergence unclear

Write a 300-word analysis: What are the trade-offs between accuracy and computational cost? Why does Active Inference typically use mean-field or Laplace rather than MCMC?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Mathematical derivation	Free energy bound
2	Analytical computation	Decompositions in Gaussian models
3	Variational updates	Mean-field optimization
4	Graphical models	Factor graph decomposition
5	Method comparison	Approximation trade-offs